ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

(i) f_x = 16/5 = 3.2 pixels, using focal length 16 mm and pixel size 5 microm. (ii) Sensor width is 6.4 mm, so field width = 400x6.4/16 = 160 mm and scale = 0.125 mm/pixel.

Q1(b) [3 marks]

ADC step = 5/4096 = 1.2207 mV. I divide by 0.1 mV/N, giving ΔF = 12.207 N.

Q1(c) [3 marks]

$$F = 0.8187, G = 0.0906$$

Question 2 [10 marks]

Q2(a) [2 marks]

The state model is $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - 0.05x_2 - 2.4525 \sin x_1)/0.02$. Equilibrium requires $\tau_0 = 2.4525 \sin \theta_e$, so τ_0 must be between -2.4525 and +2.4525 N m.

Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

Q2(c) [4 marks]

At steady state $\dot{\theta} = 0$, and I assume $\theta_{ss} = \theta_d = 90°$ because the PD controller tracks the desired angle. Thus the steady-state error is zero.

Q2(d) [2 marks]

The gravity term changes the equilibrium, so the steady-state error is still nonzero. The poles should change.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

For H(s)=ωc/(s+ωc), ωc=2π(20)=125.664 rad/s and T=0.001 s. Tustin gives H(z)=α(1+z^{-1})/[(1+α)+(α-1)z^{-1}], α=ωcT/2=0.0628319. Thus:

$$y[k] = 0.881765y[k-1] + 0.0591175(x_k + x[k-1])$$

DC gain = (2x0.0591175)/(1-0.881765) approximately 1.000.

$\omega_c = 2\pi(20) = 125.664$ rad/s. With $T = 0.001$ s, $\alpha = \omega_c T/2 = 125.664(0.001)/2 = 0.062832$. The recurrence coefficients are approximately 0.881765 on the previous output and 0.0591175 on each input sample. Check: $2(0.0591175)/(1 - 0.881765) \approx 1$, so the DC gain is close to one.

Q3(c) [4 marks]

The characteristic polynomial is s^2+(ℓ_1+2)s+(2ℓ_1+ℓ_2). I match it to s^2+10s+100 and get ℓ_1=8, ℓ_2=84.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.